Sri Lanka Institute of Information Technology

<RideLanka FleetHire>

< Web-based Car Rental Management System >

Lab Submission - 05

Sequence Diagrams

< 2026-Y2-S1-KU-36 >

Group Members and Core Ownership

No.	Member / Registration No.	Assigned Function
01	S M H C B D Rambukpitiya - IT25103736	User & customer management
02	I K S P Jayawardana - IT25103732	Vehicle management
03	W M K K G N A Weerasingha - IT25103737	Booking & reservation management
04	R M M D Bhagyamathi - IT25103733	Payment management
05	R M R D Rajapaksha - IT25103734	Feedback & support management
06	K A K Dananjaya - IT25103735	Fleet management

Software Engineering | SE2030
B.Sc. (Hons) in Information Technology

01 User & customer management Sequence Diagram

UM04 - Suspend a Customer Account for Policy Violation

OWNERSHIP AND TRACEABILITY S M H C B D Rambukpitiya - IT25103736 | Core ownership: User & customer management | Diagram use case: UM04

Use Case Recap

(Full use case specification completed in Lab Submission 04.)

Field	Detail
Use case ID	UM04
Use case name	Suspend a Customer Account for Policy Violation
Primary actor	System Administrator
Goal in context	Prevent a customer who has violated platform policy from making new bookings, while preserving an accountable status history.

Boundary / Control / Entity Classes Used

- :AccountUI (boundary) - account screen the Administrator interacts with
- :AccountManager (control) - coordinates validation, the status transition and audit recording
- :CustomerAccountDB (entity) - stores the customer account, its version and status history

UM04 Sequence Diagram



Diagram Notes

- Main flow: open record -> display status/version/history -> suspend with reason -> validate -> transition -> audit -> confirm.
- alt fragment covers guarded branches 3a (reason missing), 4a (unauthorised actor), 5a (account not found), 5b (version conflict), 6a (already SUSPENDED) and the valid-request path.
- Nested alt inside the valid-request branch covers 8a (audit/change cannot be saved) versus a successful save.

02 Vehicle management Sequence Diagram

VM03 - Mark a Vehicle Out of Service

OWNERSHIP AND TRACEABILITY I K S P Jayawardana - IT25103732 | Core ownership: Vehicle management | Diagram use case: VM03

Use Case Recap

(Full use case specification completed in Lab Submission 04.)

Field	Detail
Use case ID	VM03
Use case name	Mark a Vehicle Out of Service
Primary actor	Fleet Management Staff
Goal in context	Prevent an unsafe or unusable vehicle from receiving new bookings while preserving an accountable status history.

Boundary / Control / Entity Classes Used

- :VehicleUI (boundary) - vehicle record screen used by Fleet Management staff
- :VehicleManager (control) - coordinates validation, the status transition and audit recording
- :VehicleDB (entity) - stores the vehicle record, its version and status history

VM03 Sequence Diagram

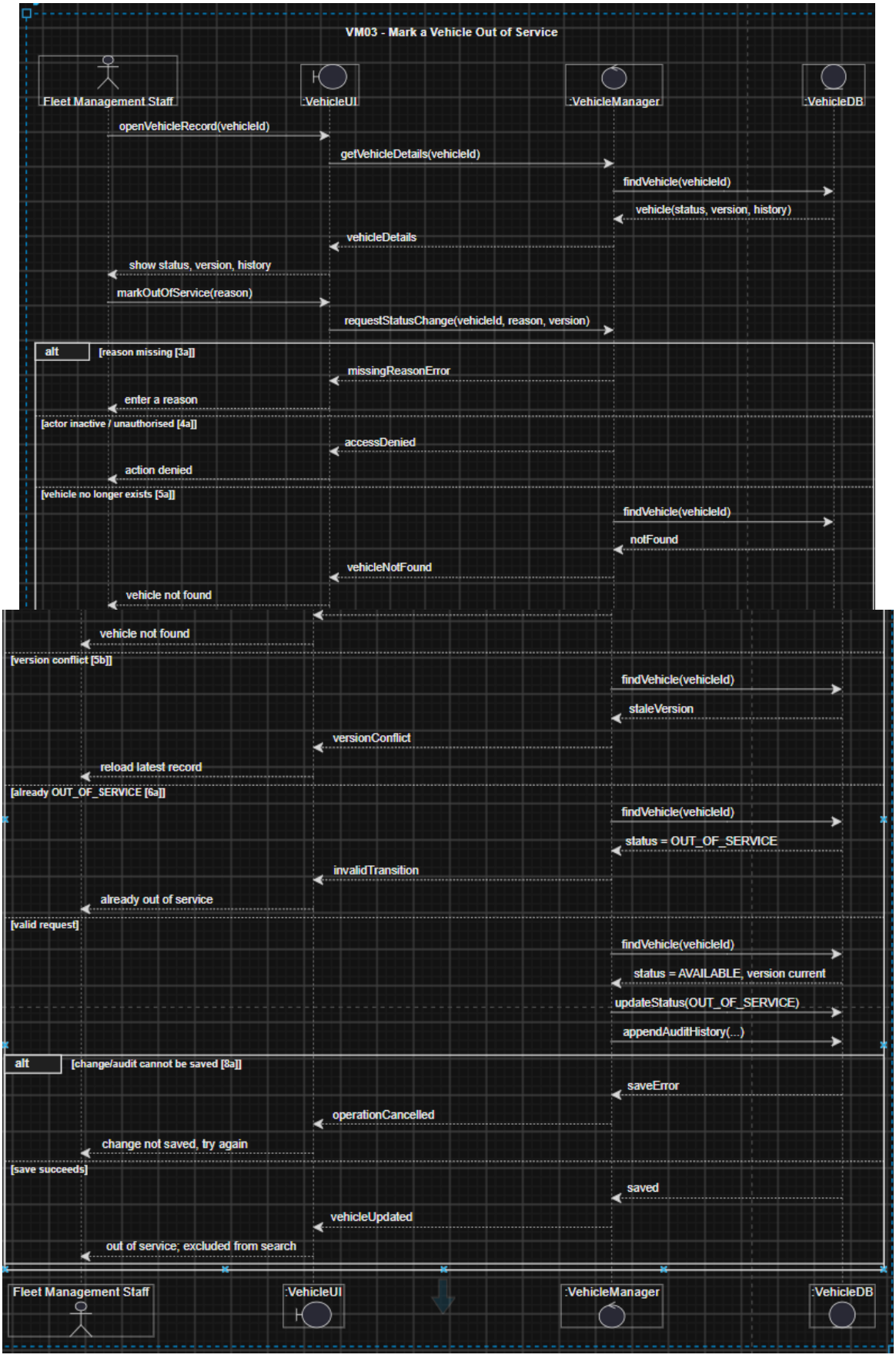


Diagram Notes

- Main flow mirrors UM04: open record -> display status/version/history -> mark out of service with reason -> validate -> transition -> audit -> confirm.
- alt fragment covers 3a (reason missing), 4a (unauthorised actor), 5a (vehicle not found), 5b (version conflict), 6a (already OUT_OF_SERVICE) and the valid-request path.
- Nested alt covers 8a (change/audit cannot be saved) versus a successful save that excludes the vehicle from search.

03 Booking & reservation management Sequence Diagram

BR04 - Reserve a Vehicle and Confirm a Booking

OWNERSHIP AND TRACEABILITY W M K K G N A Weerasingha - IT25103737 | Core ownership: Booking & reservation management | Diagram use case: BR04

Use Case Recap

(Full use case specification completed in Lab Submission 04.)

Field	Detail
Use case ID	BR04
Use case name	Reserve a Vehicle and Confirm a Booking
Primary actor	Registered Customer
Goal in context	Secure the customer's selected vehicle, complete an authenticated payment, and issue a confirmed booking without double allocation.

Boundary / Control / Entity Classes Used

- :BookingUI (boundary) - availability, hold, checkout and confirmation screens
- :BookingManager (control) - coordinates hold creation, booking creation and payment-result handling
- :VehicleDB (entity) - vehicle availability and hold state
- :BookingDB (entity) - booking record and payment outcome
- :PaymentGateway (boundary) - external payment provider
- :NotificationService (boundary) - queues the booking-confirmation message

BR04 Sequence Diagram

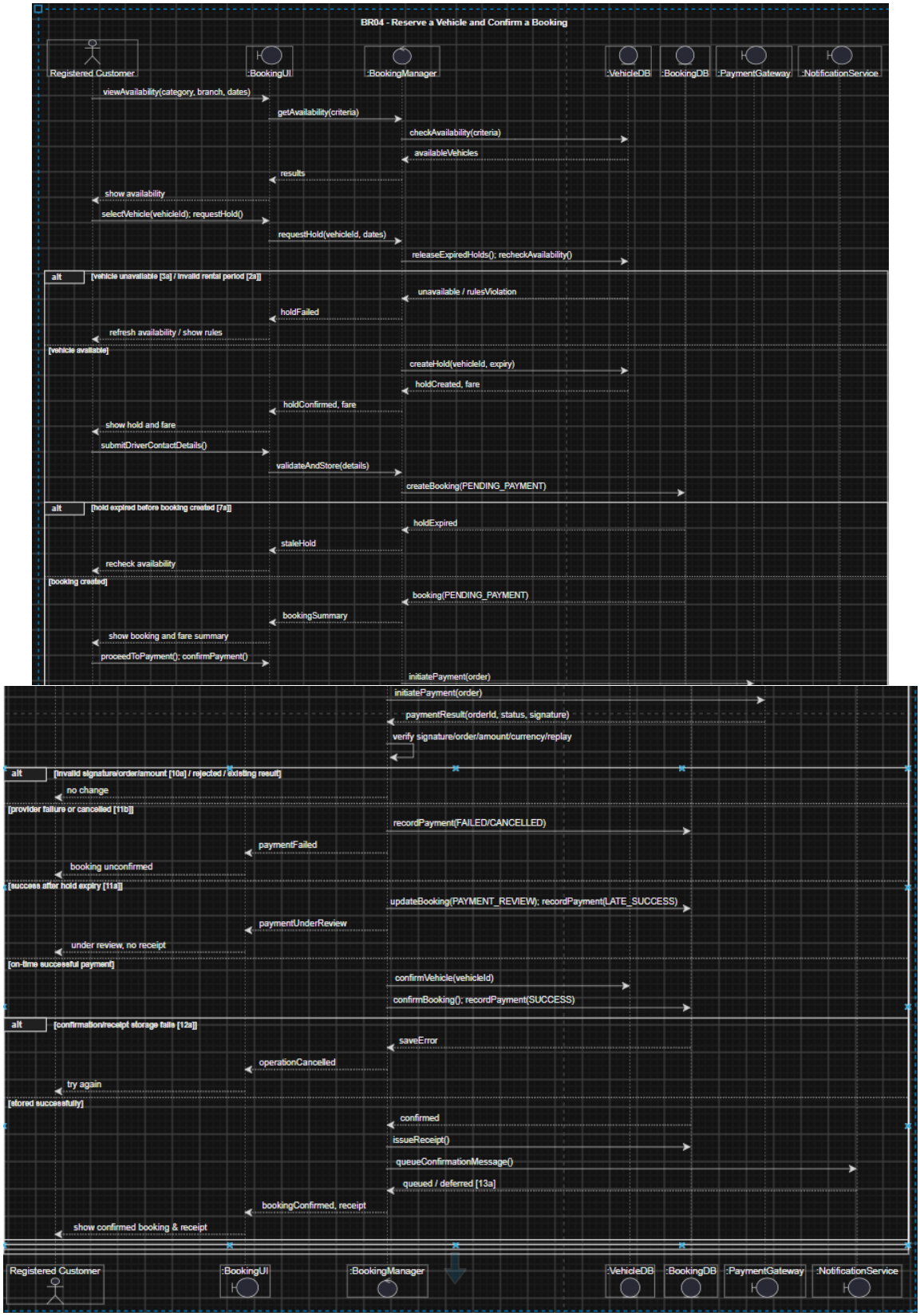


Diagram Notes

- Outer alt separates an unavailable vehicle / invalid rental period (3a, 2a) from the available-vehicle path.
- A nested alt separates a stale hold at booking time (7a) from successful booking creation.

- A further nested alt handles the payment callback: invalid/duplicate (10a, 10b), provider failure (11b), late success (11a), and on-time success.
- The innermost alt covers 12a (confirmation/receipt storage fails) versus a successful, fully confirmed booking with receipt and queued notification (13a noted on the queue response).

04 Payment management Sequence Diagram

PM03 - Verify Payment Result Before Booking Confirmation

OWNERSHIP AND TRACEABILITY R M M D Bhagyamathi - IT25103733 | Core ownership: Payment management | Diagram use case: PM03

Use Case Recap

(Full use case specification completed in Lab Submission 04.)

Field	Detail
Use case ID	PM03
Use case name	Verify Payment Result Before Booking Confirmation
Primary actor	Payment Gateway
Goal in context	Accept a genuine payment-gateway result exactly once and prevent booking confirmation for an invalid, mismatched, failed, or late payment.

Boundary / Control / Entity Classes Used

- :PaymentCallbackAPI (boundary) - receives the gateway callback
- :PaymentManager (control) - validates, maps outcomes and coordinates booking confirmation
- :PaymentDB (entity) - payment state and retained callback evidence
- :BookingDB (entity) - booking and hold state consulted for the match check
- :VehicleDB (entity) - vehicle hold validity check

PM03 Sequence Diagram

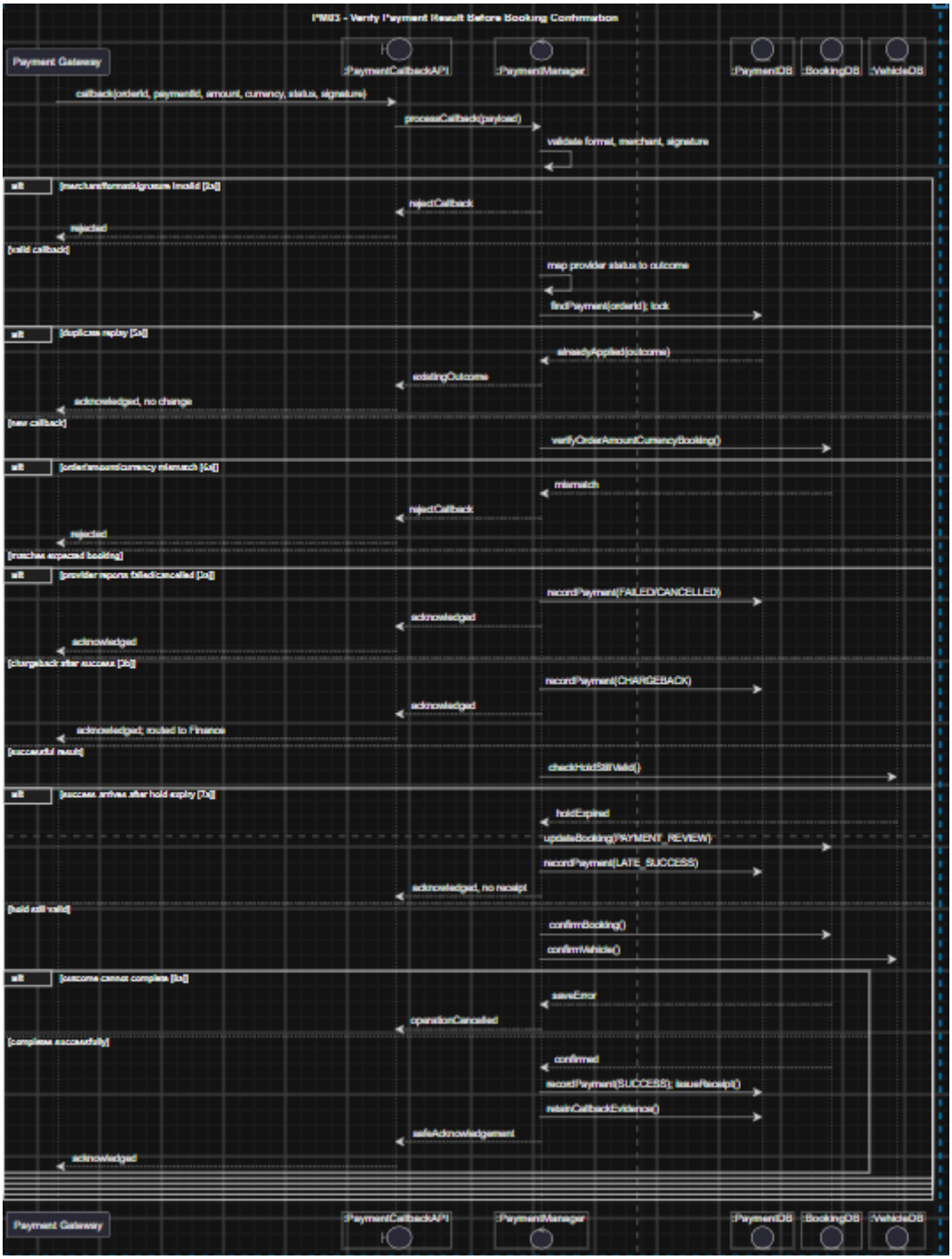


Diagram Notes

- Outer alt: 2a (invalid merchant/format/signature) versus a valid callback.
- Nested alt: 5a (duplicate replay) versus a new callback needing the order/amount/currency/booking match check (6a mismatch handled in the next alt).
- Further nested alt distinguishes 3a (failed/cancelled), 3b (chargeback after success), and a successful result.
- Innermost alt: 7a (success after hold expiry -> PAYMENT_REVIEW / LATE_SUCCESS) versus a valid hold, then 8a (cannot complete) versus a fully completed, receipted confirmation.

05 Feedback & support management Sequence Diagram

FS02 - Submit and Assign a Customer Complaint Ticket

OWNERSHIP AND TRACEABILITY R M R D Rajapaksha - IT25103734 | Core ownership: Feedback & support management | Diagram use case: FS02

Use Case Recap

(Full use case specification completed in Lab Submission 04.)

Field	Detail
Use case ID	FS02
Use case name	Submit and Assign a Customer Complaint Ticket
Primary actor	Registered Customer
Goal in context	Capture a customer complaint accurately and route it to the correct staff member for timely resolution.

Boundary / Control / Entity Classes Used

- :ComplaintUI (boundary) - booking history and complaint-submission screen
- :TicketManager (control) - validates the request, creates the ticket and determines routing
- :BookingDB (entity) - verifies booking ownership
- :TicketDB (entity) - stores the ticket, queue assignment and activity log

FS02 Sequence Diagram

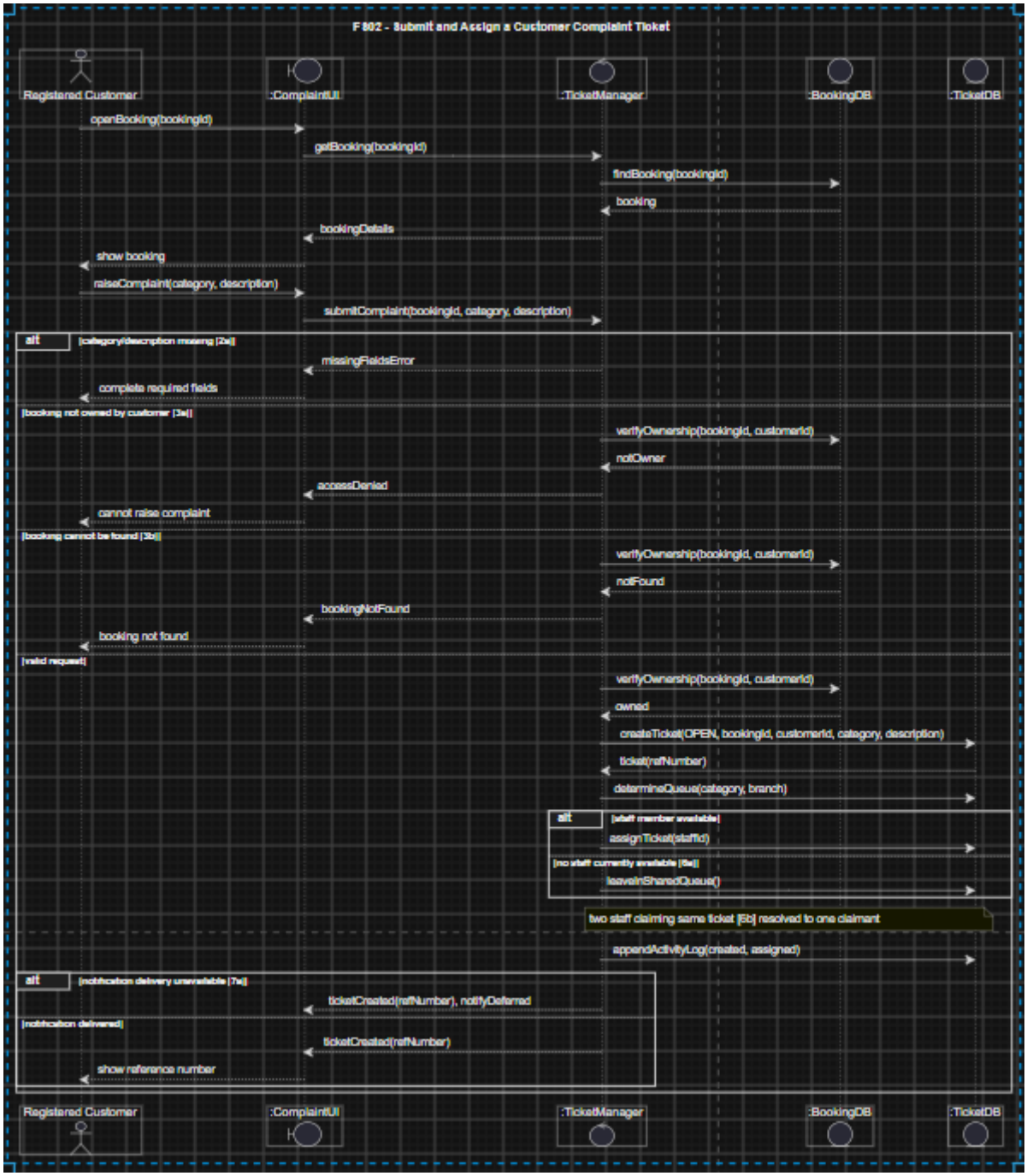


Diagram Notes

- alt fragment covers 2a (missing category/description), 3a (booking not owned by customer), 3b (booking not found), and the valid-request path.
- Within the valid-request path, a nested alt shows staff-available assignment versus 6a (no staff available, left in shared queue); a note documents 6b (two staff claiming the same ticket resolved to one claimant).
- A further alt covers 7a (notification delivery temporarily unavailable) versus successful notification, before the reference number is shown to the customer.

06 Fleet management Sequence Diagram

FM03 - Schedule and Complete Vehicle Maintenance

OWNERSHIP AND TRACEABILITY K A K Dananjaya - IT25103735 | Core ownership: Fleet management | Diagram use case: FM03

Use Case Recap

(Full use case specification completed in Lab Submission 04.)

Field	Detail
Use case ID	FM03
Use case name	Schedule and Complete Vehicle Maintenance
Primary actor	Maintenance Coordinator
Goal in context	Plan a vehicle's maintenance window, temporarily remove it from booking eligibility, and record the completed service history.

Boundary / Control / Entity Classes Used

- :MaintenanceUI (boundary) - maintenance calendar and completion screens
- :MaintenanceManager (control) - coordinates conflict checks, scheduling and completion
- :BookingDB (entity) - confirmed bookings consulted for conflicts
- :MaintenanceDB (entity) - maintenance record and service history
- :VehicleDB (entity) - vehicle status and availability

FM03 Sequence Diagram



Diagram Notes

- Outer alt: 3a (service type/notes missing), 4a (window conflicts with a confirmed booking), and the no-conflict path.
- Within the no-conflict path, a nested alt handles 5a (vehicle already OUT_OF_SERVICE) versus blocking bookings for an available vehicle.
- A further alt covers 6a (attempt to start before the window begins) versus a valid IN_PROGRESS transition.
- The final alt covers 7a (incomplete completion details) then a nested alt for 8a (record cannot be saved) versus a saved COMPLETED record that restores vehicle availability.